

MEHDI DOSS

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EDUCATION

ENSI - National School of Computer Science, Manouba, Tunisia 2024 - 2027
Engineering Student in Computer Science

FSM - Faculty of Sciences of Monastir, Monastir, Tunisia 2022 - 2024
Preparatory Studies in Mathematics and Physics – Ranked 350th out of 1,700 in the National Engineering Exam

EXPERIENCE

Summer Intern – Artificial Intelligence Jul 2026 – Aug 2026
Hutchinson *Sousse, Tunisia*

- Developed a cross-platform real-time production monitoring system using Tauri (Rust), TypeScript, Vite, Hono, and Bun, with WebSocket-based data visualization.
- Integrated an LLM-powered assistant to analyze live industrial data, detect anomalies, generate summaries, and answer natural-language queries.

Summer Intern – Artificial Intelligence Aug 2025
SWConsulting *Monastir, Tunisia*

- Loan Application Prediction:** Built an end-to-end loan approval prediction pipeline using scikit-learn, benchmarking 5+ machine learning models before selecting TabPFN for a 92% prediction accuracy.
- Deployed the solution with Streamlit for interactive real-time inference.

TECHNICAL SKILLS

Machine Learning	Deep Learning, Computer Vision, CNN, YOLO, RAG
LLM & Generative AI	LangGraph, Prompt Engineering, Vector Databases
Programming	Python, C++, JavaScript
ML Frameworks	PyTorch, TensorFlow, Scikit-Learn, OpenCV
Data & Analysis	NumPy, Pandas, Matplotlib, Seaborn, Data Preprocessing
Model Development	Hyperparameter Tuning, Cross-Validation, Model Optimization
Web Development	React, Flutter, HTML, CSS, Vite, Tauri, Node.js, Hono, Bun, REST APIs
Deployment & MLOps	FastAPI, Docker, CI/CD, Git, GitHub, GitLab, Streamlit
Databases	MongoDB, SQL
Embedded & Edge AI	Raspberry Pi, ESP32
Languages	Arabic (Native), English (B2), French (B2), German (A1)
Personal Skills	Leadership, Teamwork, Public Speaking

PROJECTS

CropEye – Autonomous Agricultural Drone: Engineered a real-time olive tree disease detection system by training a **YOLOv8n** object detection model to classify healthy vs. Verticillium wilt-affected trees, reaching **0.80 mAP@0.5**, **0.95 AP** on the healthy class, **83% precision**, and **81% recall** on a held-out test set of **500** images. Streamlined the model for edge inference by exporting to ONNX and applying **INT8** post-training quantization, reaching **8–12 FPS** on a Raspberry Pi 4 at 320px resolution. Built a Model Context Protocol (MCP) autonomous control layer enabling natural-language flight-mission planning and execution without manual piloting.

Plant Disease Classifier – Edge-Optimized Computer Vision: Delivered **94.35%** Top-1 accuracy by training and fine-tuning a MobileNetV2 classifier on the PlantVillage dataset, then compressed it with TensorFlow Lite INT8 quantization for edge deployment, limiting the accuracy drop to **1.6%** (96.4% → 94.8%) while significantly reducing model size.

PlateVision – License Plate Recognition System: Reached **90%+** recognition accuracy by engineering an AI-powered license plate recognition pipeline (YOLO-based detection + EasyOCR), with automatic validation logic and real-time vehicle tracking for continuous unattended monitoring.

CERTIFICATIONS

- **Supervised Machine Learning: Regression and Classification** – DeepLearning.AI / Stanford (Coursera)
- **Understanding Machine Learning** – DataCamp
- **Supervised Learning with Scikit-Learn** – DataCamp
- **Introduction to Deep Learning with PyTorch** – DataCamp

ACHIEVEMENTS & LEADERSHIP

- **Expert Member – Association Robotique ENSI (ARE):** Served as an expert member and trainer during RoboDay workshops, supporting students in embedded systems and robotics fundamentals.